

Day 2: Conditional Statements

if, if-else, else-if ladder, and nested if in C++

2. Conditional Statements: if, if-else, and Nested if

2.1 What is a condition?

A condition is an expression that evaluates to either true or false. A conditional statement allows a program to make a decision based on the input.

For example: If marks are greater than or equal to 40, the student passes. Otherwise, the student fails.

Types of conditions

A. Simple if

Executes a block only when the condition is true.

```
if (marks >= 40)
{
    cout << "Pass";
}
```

Use when an action is required only if a condition is satisfied.

B. if-else

Selects one of two alternatives.

```
if (age >= 18)
{
    cout << "Adult";
}
else
{
    cout << "Minor";
}
```

Use when there are two possible outcomes.

C. else-if ladder

Checks multiple conditions in sequence.

```
if (marks >= 90)
    cout << "A";
else if (marks >= 75)
    cout << "B";
else
    cout << "C";
```

Use when selecting from multiple categories.

D. Nested if

Places one condition inside another.

```

if (age >= 18)
{
    if (hasID == 1)
        cout << "Entry allowed";
}

```

Use when a second decision depends on the first decision.

Important operators

Operator	Meaning	Example
==	Equal to	a == b
!=	Not equal to	a != b
>	Greater than	a > b
<	Less than	a < b
>=	Greater than or equal to	marks >= 40
&&	AND (both conditions true)	age >= 18 && age <= 60
	OR (at least one true)	a == 0 b == 0

Remember: = assigns a value, while == compares two values.

Practice 2: Three conditional programs

Example 1: Check whether a number is even or odd

Problem: Accept an integer and determine whether it is even or odd.

Logic: A number is even if its remainder after division by 2 is zero.

```

#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    if (num % 2 == 0)
    {
        cout << "Even number";
    }
    else
    {
        cout << "Odd number";
    }

    return 0;
}

```

Sample input: 17

Output:

```
Odd number
```

Where to apply: Checking even/odd array indexes, identifying alternate positions, and validating numeric conditions.

Example 2: Find the largest of three numbers

Problem: Accept three numbers and display the largest.

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c;

    cout << "Enter three numbers: ";
    cin >> a >> b >> c;

    if (a >= b && a >= c)
    {
        cout << "Largest = " << a;
    }
    else if (b >= a && b >= c)
    {
        cout << "Largest = " << b;
    }
    else
    {
        cout << "Largest = " << c;
    }

    return 0;
}
```

Sample input: 25 40 15

Output:

```
Largest = 40
```

Notice that && ensures that both comparisons are true before selecting a number as the largest.

Where to apply: Finding maximum values, selecting the highest marks, and comparing node values in a tree.

Example 3: Check eligibility using nested if

Problem: A student can appear for an examination if attendance is at least 75%. If attendance is sufficient, check whether the student has paid the examination fee.

```
#include <iostream>
using namespace std;

int main()
{
    float attendance;
    int feePaid;

    cout << "Enter attendance percentage: ";
    cin >> attendance;

    if (attendance >= 75)
    {
        cout << "Has fee been paid? (1=Yes, 0=No): ";
        cin >> feePaid;

        if (feePaid == 1)
        {
            cout << "Eligible for examination";
        }
        else
        {
            cout << "Pay examination fee";
        }
    }
    else
    {
        cout << "Insufficient attendance";
    }

    return 0;
}
```

Sample input:

```
Enter attendance percentage: 80
Has fee been paid? (1=Yes, 0=No): 1
```

Output:

```
Eligible for examination
```

Where to apply: Menu validation, checking whether a graph vertex exists before accessing it, and checking whether a tree is empty before searching.

Practice Problems

Try solving these on your own using the conditional statements covered above.

Problem 1: Student Scholarship Eligibility System

Problem Statement

A college wants to automatically determine whether a student is eligible for a scholarship.

Accept the following information from the user:

- Student marks percentage

- Attendance percentage
- Annual family income
- Number of extracurricular activities completed

Problem 2: ATM Transaction Validation

Problem Statement

Develop a simple ATM transaction validation program.

Accept:

- Account balance
- Withdrawal amount
- PIN
- Account type (1 = Savings, 2 = Current)
- Whether the ATM has sufficient cash (1 = Yes, 0 = No)